

# **Basic Method and Plan**

**Using UVA approach**

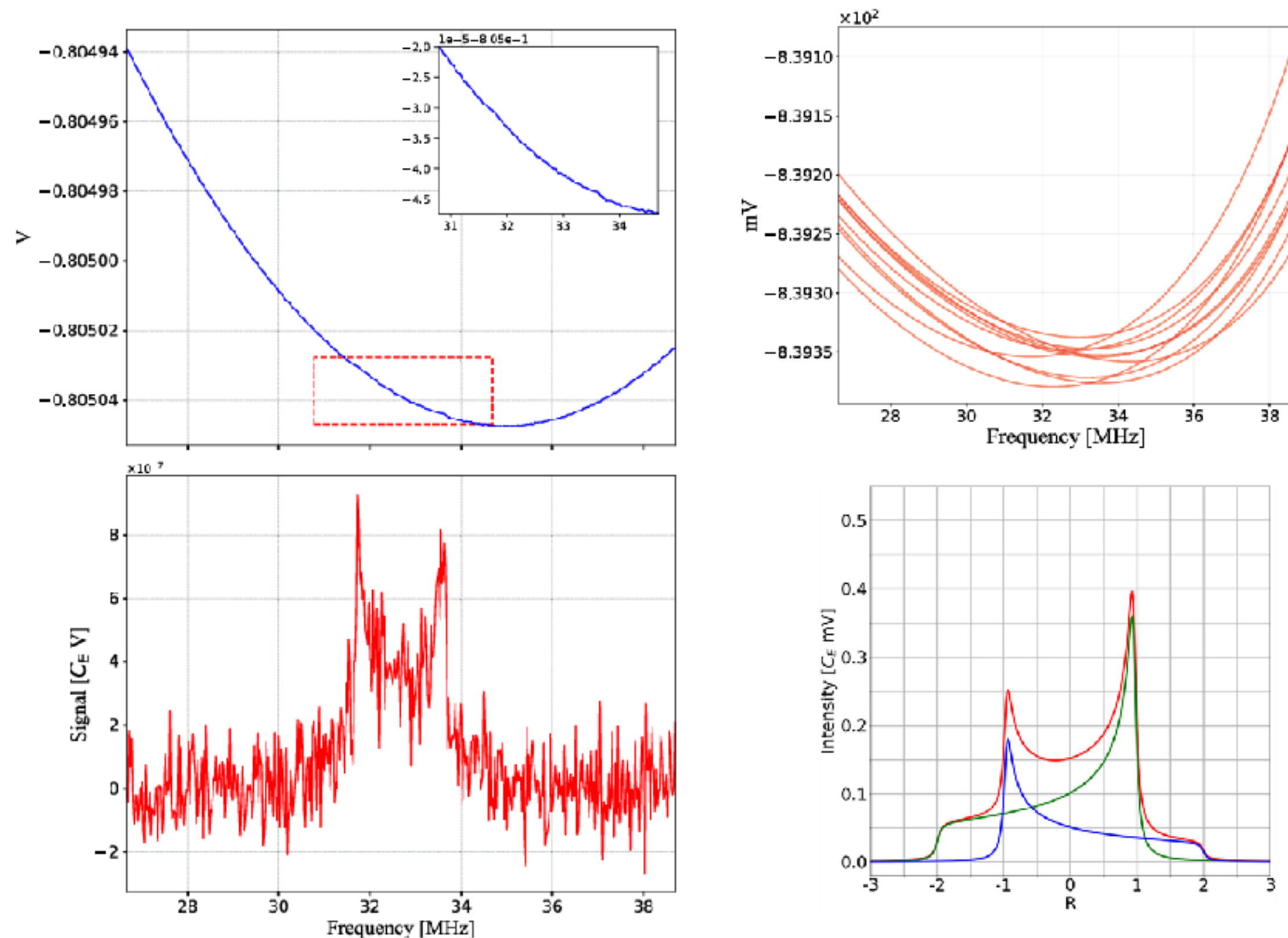
**Dustin Keller 9/1/26**

# Contents

- Foundation of simulations: basis vs tensor manipulated
- Simulation Based Inference (and why we need it)
- Calibrate and match simulations to experimental data
- Benchmarks, metrics and other uncertainties (with complexity)
- Deployment

# Initial Basis Models (No RF Manipulation)

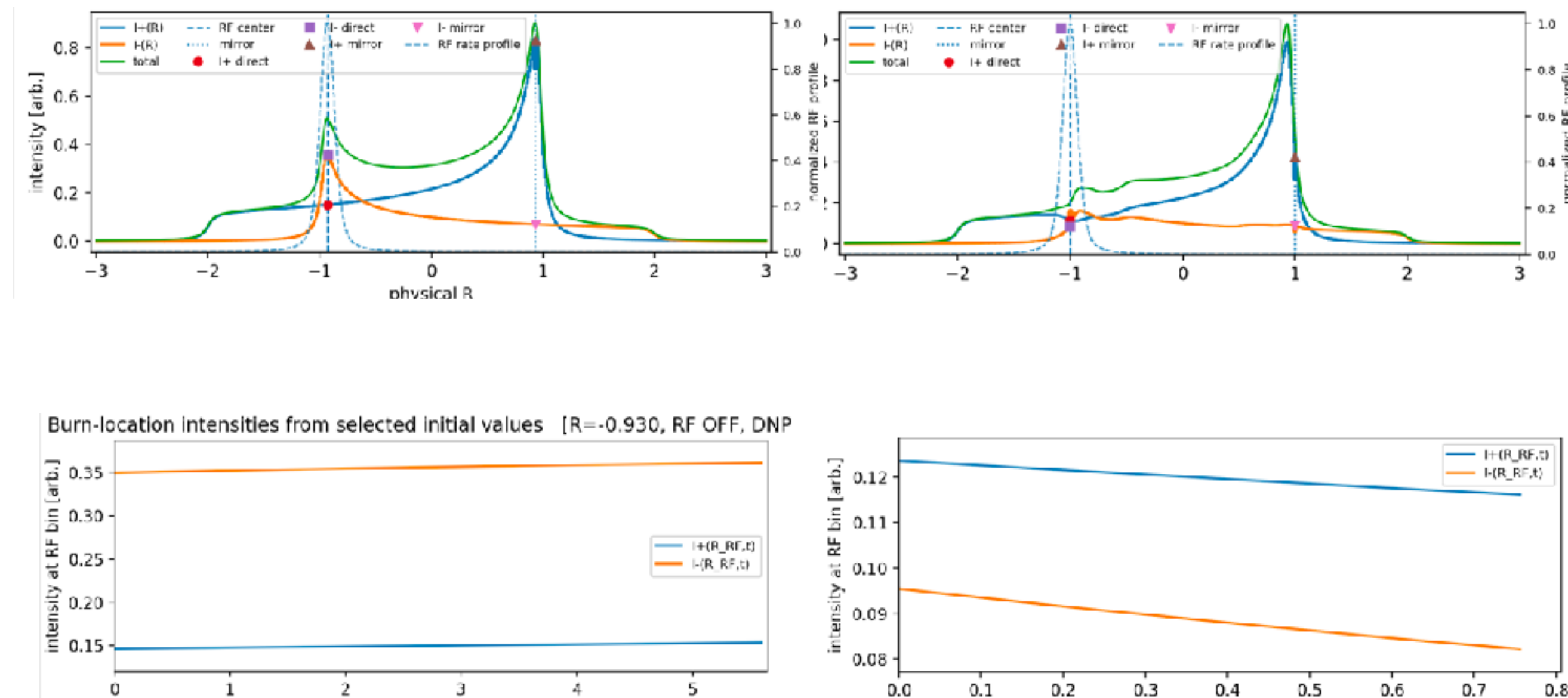
Better than fitting but only if very high quality MC is matched and calibrated



- Starting with Q-meter circuit theory models of the baseline (Q-meter tune, cable character, coil specifics)
- Add signal based on standard lineshape theory of the deuteron, but highly tune to experimental configurations (material characteristics, scales)
- Add noise: structure and scale paying attention to both relative and absolute contributions
- The methodology requires being able to reproduce the experimental data but also vary all these components we can control to make the model more adaptive and robust.

# Initial Basis Models (No RF Manipulation)

Better than fitting but only if very high quality MC is matched and calibrated



- Along with NMR spectrum you also have  $\Gamma(\text{power})$  and  $t_{\text{RF}}(\text{time applied})$ . Only these two new features
- Generate new data within selected domain and train
- Ideally every bin has a different  $\Gamma$  and  $t$  to shape the overlapping transitions as needed to fully tensor enhance.
- Greater power is needed only quickly and at select locations.

- Same as basis model by instead of basic lineshape theory you have tensor manipulated lineshape theory
- The physics model can be very complex and involved depending on target material so we generalize the dependencies so it can be calibrated to real target material
- Unlike the basis model this is a dynamic calibration: burns of different width and coils Q-factor over the frequency domain both full and partial holes and full recovery
- Obtaining very narrow burns and recovery is critical of calibration as it helps to scale relative evolution as a function of spin density
- Broad burns help to calibrate spin diffusion components
- We chose a narrow domain of the power profile and pulse timing to prevent degeneracies in the overlapping absorption line forcing single solution per burn

# Characterize Spin Diffusion

## With and without DNP

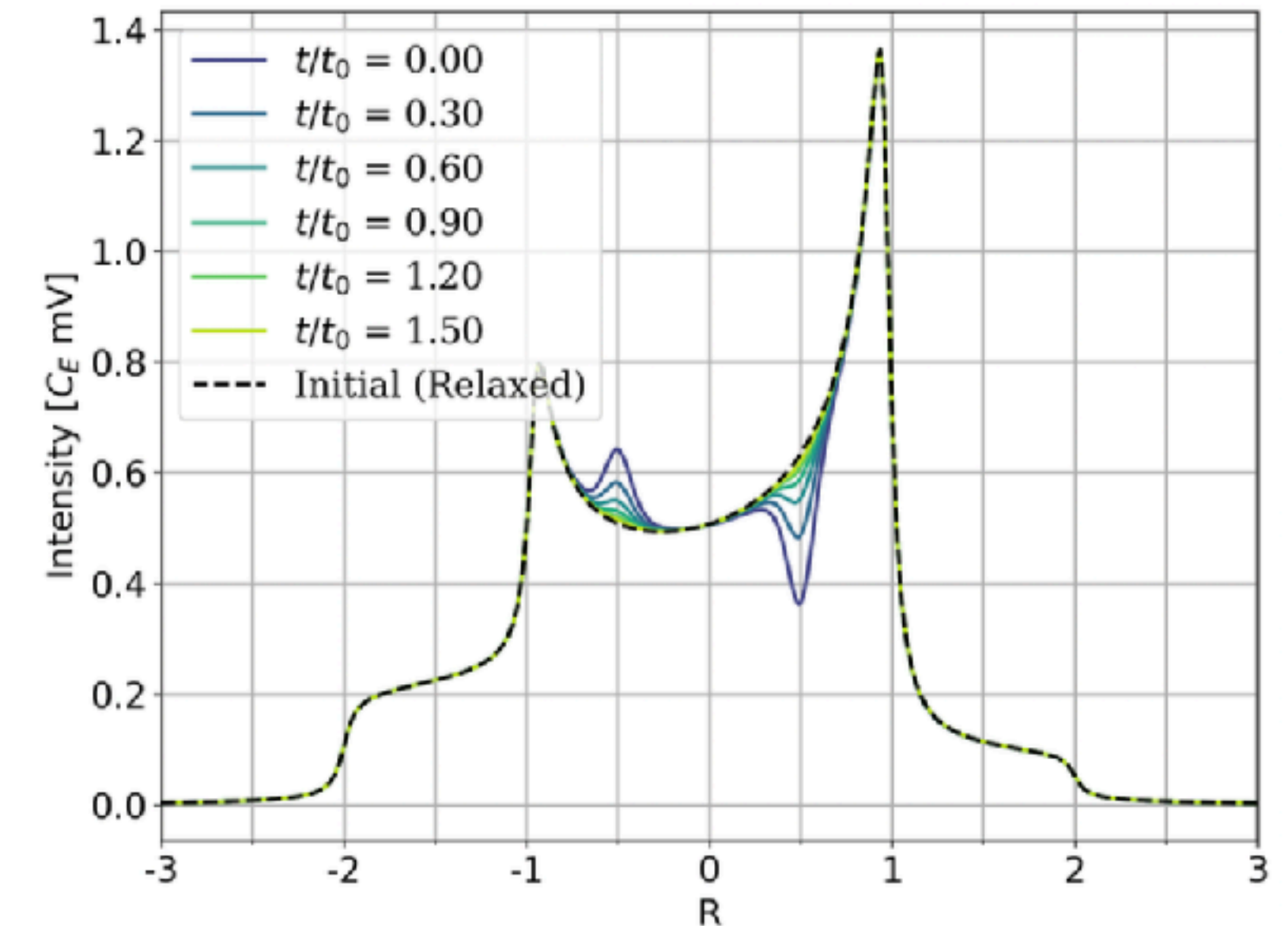
- Assume diffusion changes not only as a function of intensity but also the angle.
- Then map out from measurements at different intensity and angle to quantify.
- Stick it in the expression and test by predicting recovery under not measured scenarios.

$$\frac{dn_i(\theta_0, \vec{r})}{dt} \supset D[n_i(\theta_0, \vec{r})] \nabla^2 n_i(\theta_0, \vec{r})$$

the variable  $\theta_0$  labels a *fixed subset of spins* whose internuclear (or EFG) axis makes an angle  $\theta_0$  with the external field.

The diffusion operator  $\nabla^2$  here acts in **real space** (the spatial coordinate  $\vec{r}$ ), describing **spatial spin diffusion** — i.e. polarization exchange between nearby sites in the material through dipolar flip-flop processes.

<https://doi.org/10.1140/epja/s10050-026-01790-y>



Continuum Laplace–Beltrami form

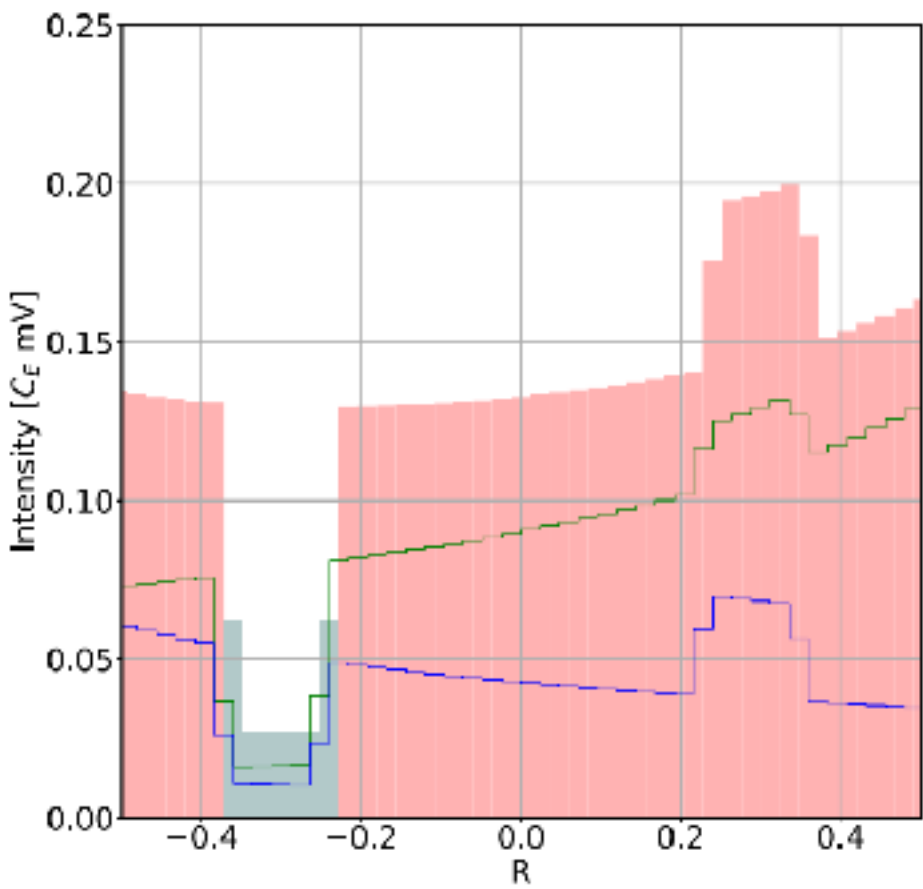
$$\left( \frac{\partial n_i}{\partial t} \right)_{\text{diff}} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( D(\theta) \sin \theta \frac{\partial n_i}{\partial \theta} \right)$$

$\theta$  is orientation of the coupled spin packets along the NMR line

# Error from HP-limit

## Acquired bin by bin

| Target error | Required pulse duration at $f_b$           |
|--------------|--------------------------------------------|
| 1%           | $t_b \lesssim 0.01 \tau_{\text{dyn}}(f_b)$ |
| 5%           | $t_b \lesssim 0.05 \tau_{\text{dyn}}(f_b)$ |
| 10%          | $t_b \lesssim 0.10 \tau_{\text{dyn}}(f_b)$ |



$$\epsilon_{\text{sd/evol}}(t_b, f_b) = \eta_Q(f_b) \left[ 1 - e^{-t_b/\tau_{\text{dyn}}(f_b)} \right]$$

*error due to signal evolution*

$$\tau_{\text{dyn}}(f_b) = \left[ \frac{1}{\tau_{\text{diff}}(f_b)} + \frac{1}{T_1(f_b)} \right]^{-1}$$

*dynamic time at frequency bin*

$$t_b \lesssim -\tau_{\text{dyn}}(f_b) \ln \left[ 1 - \frac{\epsilon_{\text{tol}}}{\eta_Q(f_b)} \right]$$

*High power limit time threshold*

$$\epsilon_{\text{sd/evol}}(t_b, f_b) \approx \eta_Q(f_b) \frac{t_b}{\tau_{\text{dyn}}(f_b)} \qquad t_b \ll \tau_{\text{dyn}}(f_b)$$

*contribution from over-burn*

$$\epsilon_{\text{center}}(t_b, f_b) = \frac{|a_c(t_b, f_b) - a_{\text{tar}}(f_b)|}{a_{\text{tar}}(f_b)}$$

*Total error at that bin*

$$\epsilon_{\text{total}}(t_b, f_b) \approx \epsilon_{\text{center}}(t_b, f_b) + \eta_Q(f_b) \left[ 1 - e^{-t_b/\tau_Q(f_b)} \right]$$

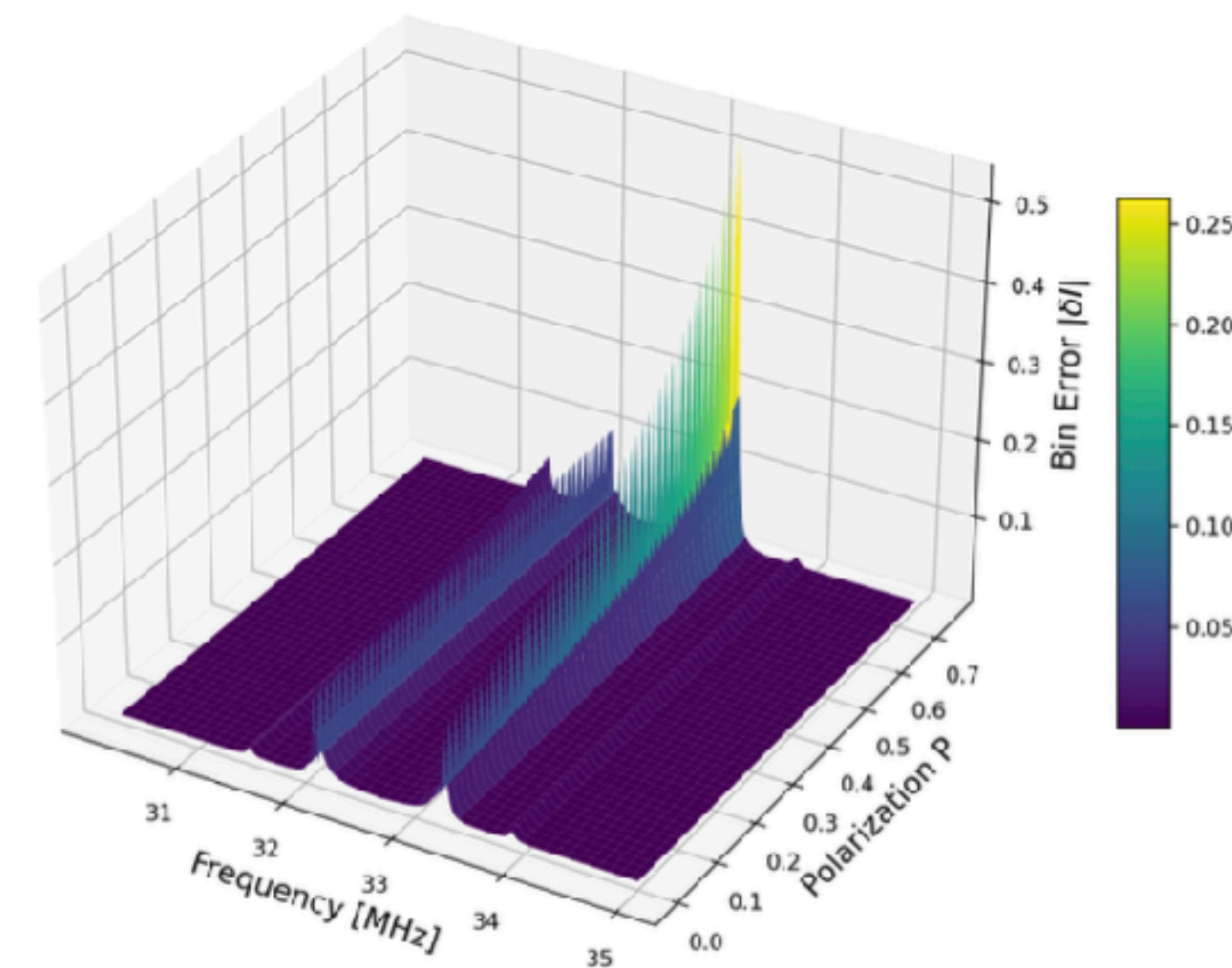
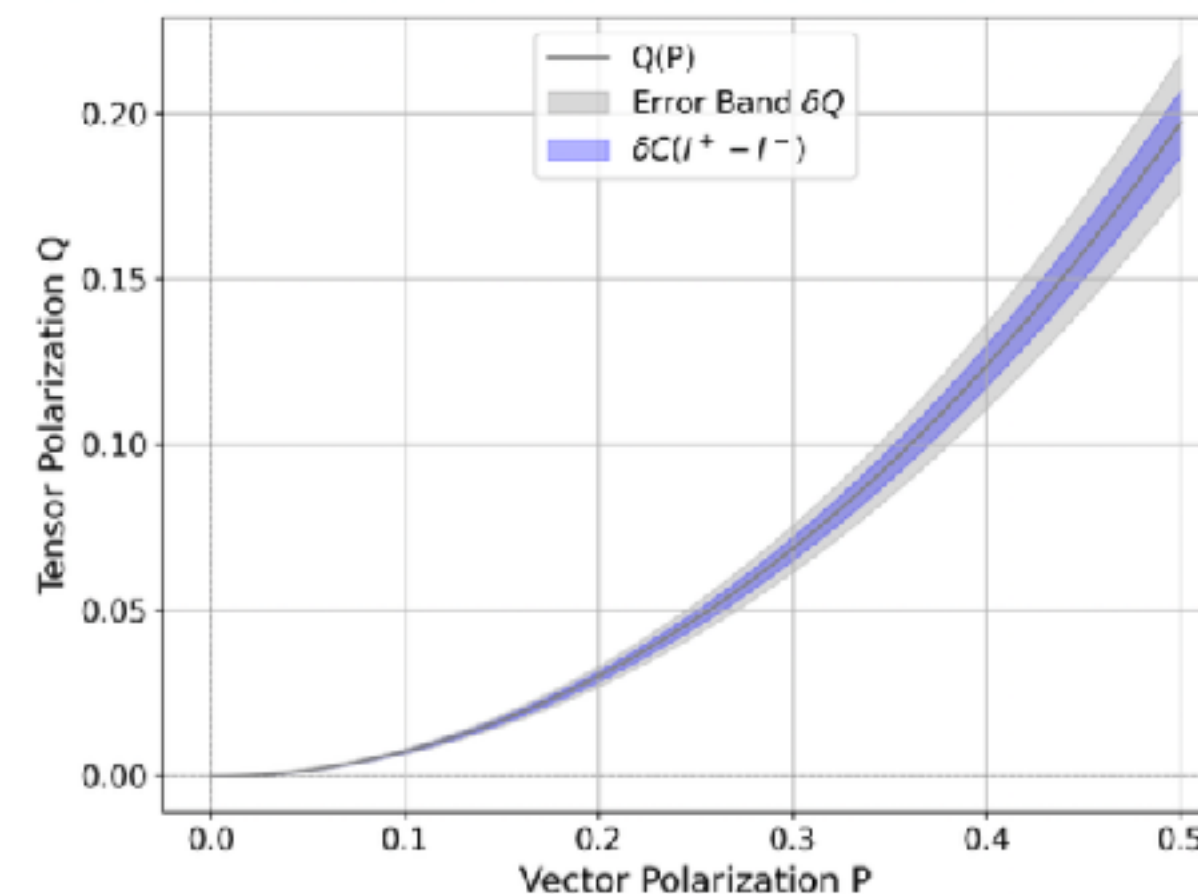
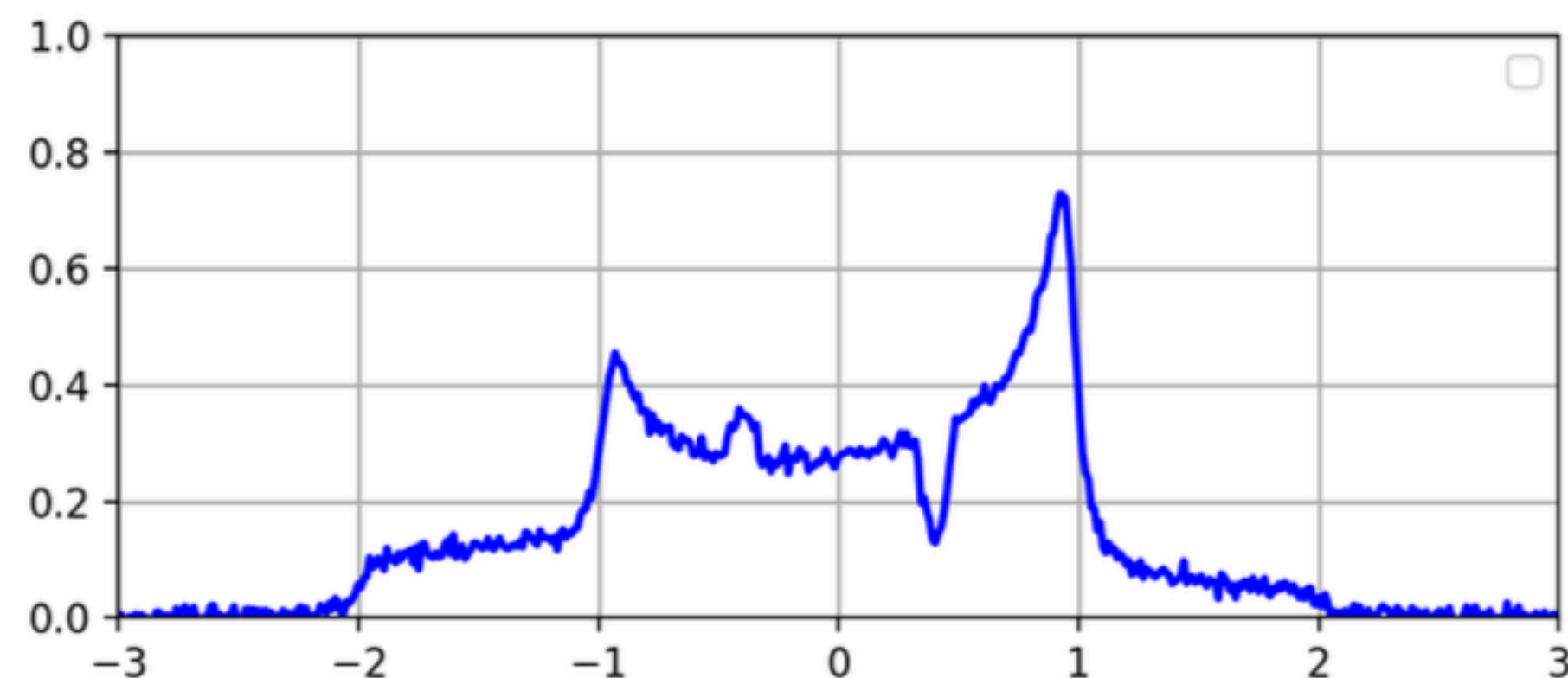
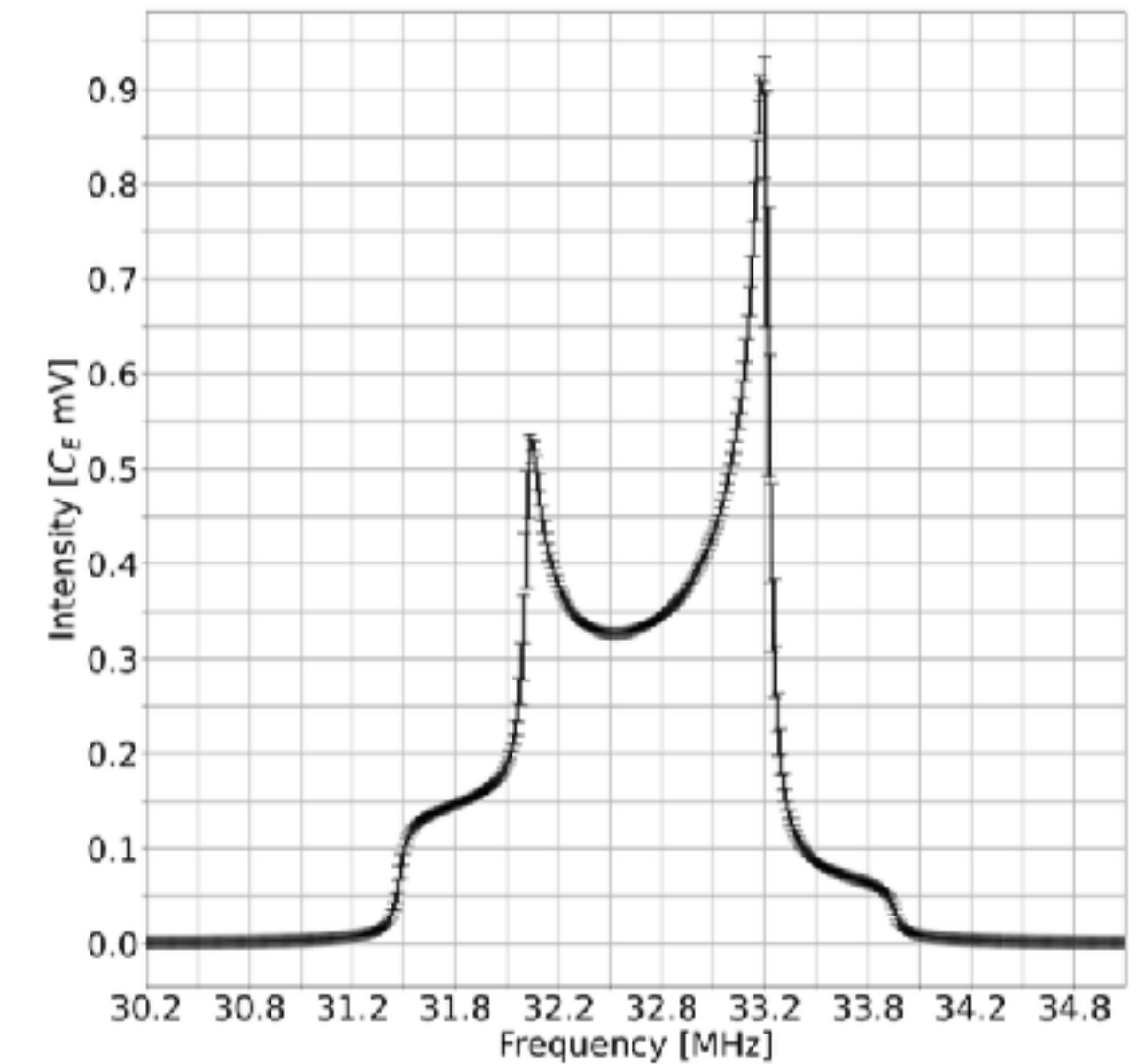
*Error from HP-limit is only induced at the limit*

| Burn time | error  | Practical interpretation                |
|-----------|--------|-----------------------------------------|
| 0.05 s    | 0.10%  | Safely high-power / pulse-like          |
| 0.10 s    | 0.20%  | Safely high-power / pulse-like          |
| 0.20 s    | 0.40%  | Very small correction                   |
| 0.30 s    | 0.60%  | Still essentially negligible            |
| 0.50 s    | 1.00%  | Chosen high-power cutoff                |
| 0.75 s    | 1.50%  | Small but no longer zero                |
| 1.00 s    | 1.99%  | Correction should be tracked            |
| 1.50 s    | 2.97%  | Noticeable correction                   |
| 2.00 s    | 3.94%  | Not negligible; usable with correction  |
| 3.00 s    | 5.85%  | Significant dynamic error               |
| 5.00 s    | 9.56%  | Long-burn regime                        |
| 10.00 s   | 18.21% | Strongly model-dependent                |
| 20.00 s   | 33.10% | Approaching driven-equilibrium behavior |

# How To Make Sense out of NMR data

## Error specific to ss-RF

- Bin Error (from differential binning) - smaller bins...
- Changes in signal during NMR (sensitive to sweep rate)
- Averaging over sweeps is greater error than not
- Burning bin error (lineshape and evolution)



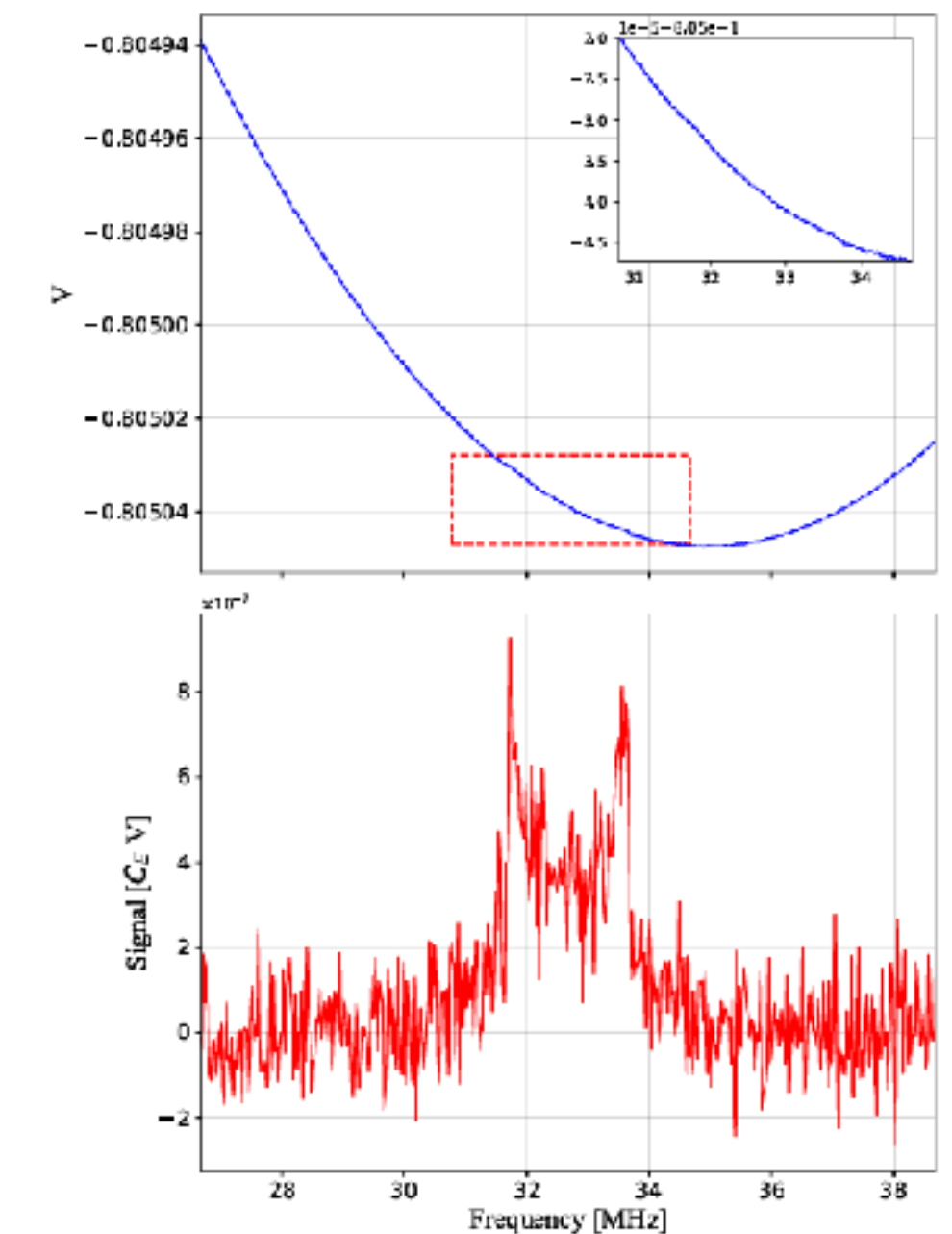
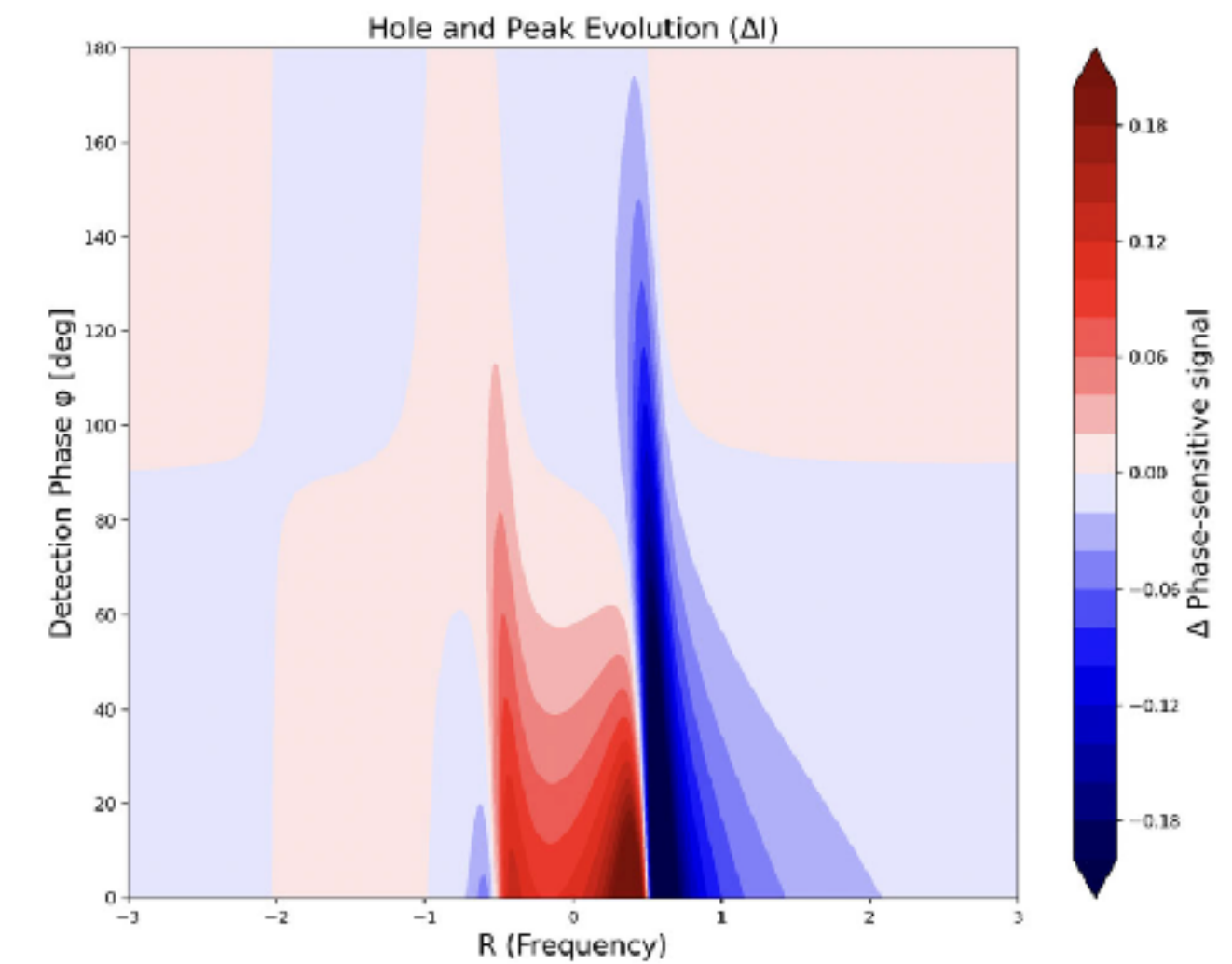
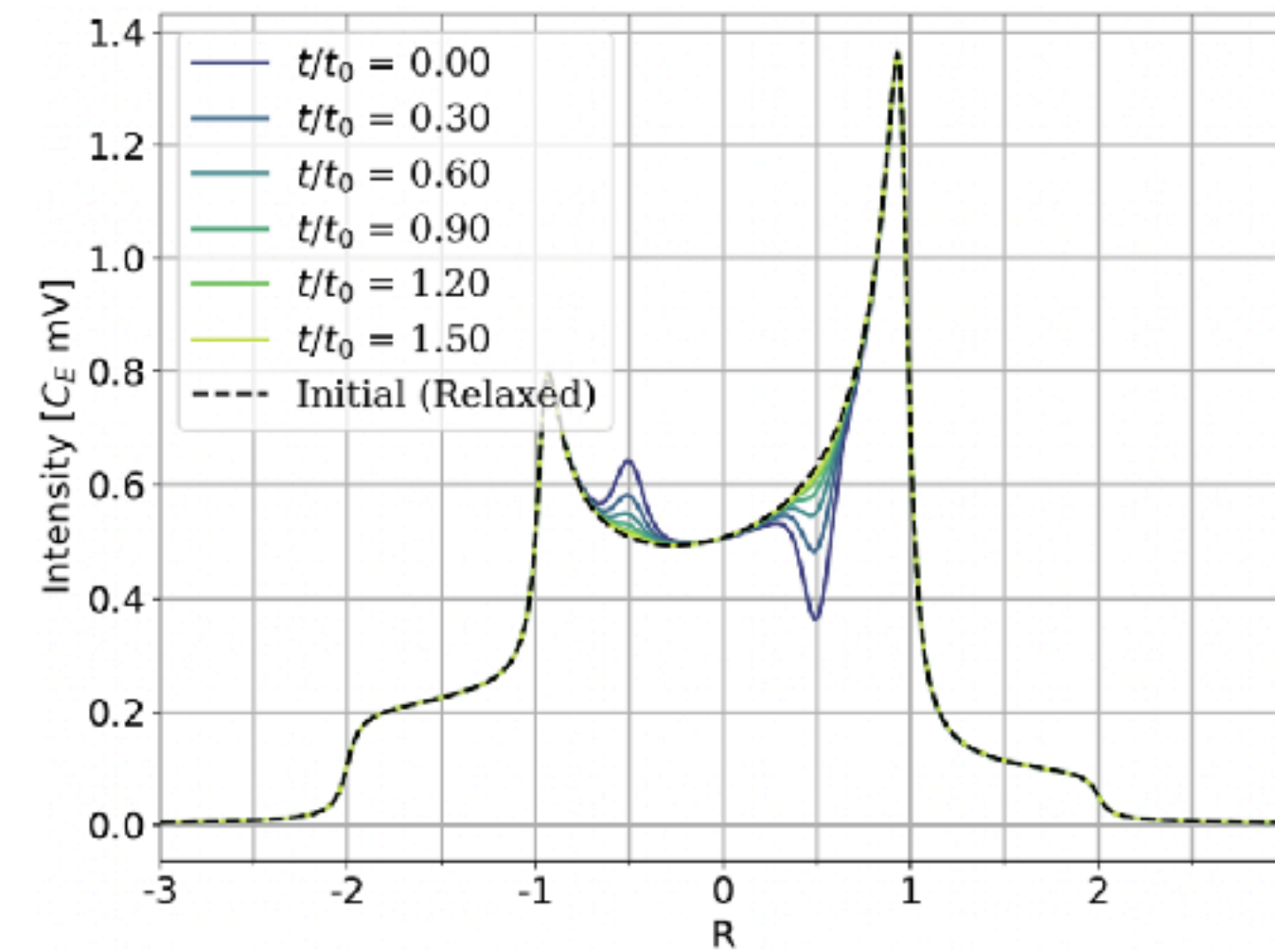
# Resolving Bin Errors

With more...

- More information
- More bins
- Multiple phase measurements
- More time steps
- Turn into time sequenced data
- Many larger error time steps
- Much more signal information
- Much more evolution details
- CNN...

Sticking to 1D for this project

But we are working on 2D/3D



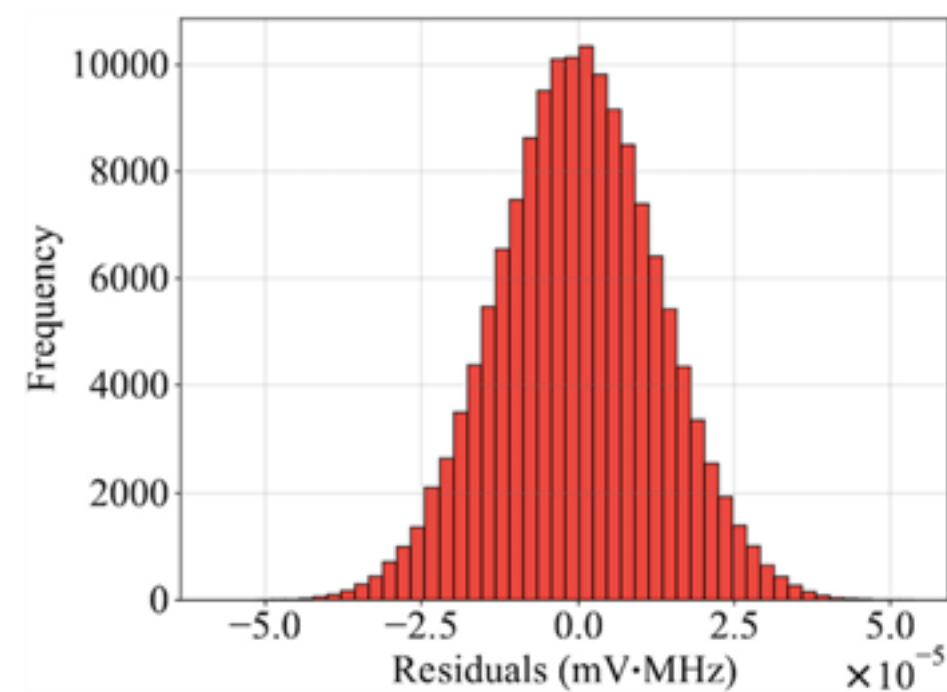
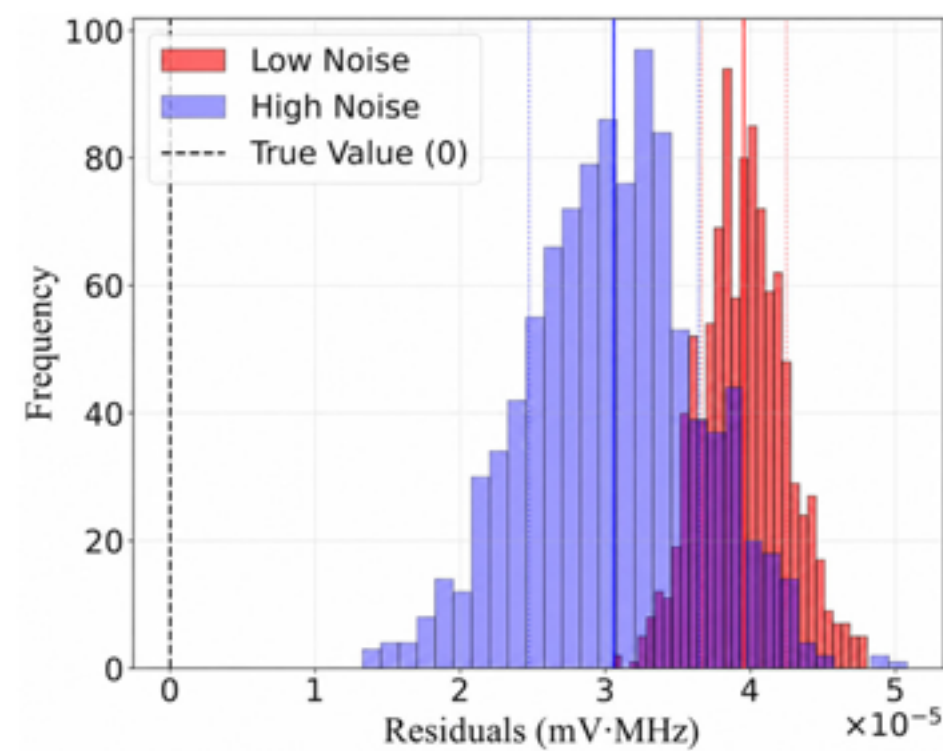
# Calibration and Matching

## Specifically for ND3

- Use what data we have (UVA, UHN, JLab)
- Need both static and dynamic calibration
  - Static - basic lineshape and baseline environment
  - Dynamic - full program of data taking to parameterized ND3 evolution under DNP, no-DNP, different thermal conditions, radiation conditions, spin diffusion in wide and narrow burns.
- Integrate and focus on 5-55% vector with 1-35% tensor

# Benchmarks, Metrics and Uncertainty

## As defined



$$\sigma = \sqrt{\frac{1}{N} \sum_{n=1}^N (\Delta P_n - \mu)^2}.$$

$$\mu = \frac{1}{N} \sum_{n=1}^N \Delta P_n,$$

$$\epsilon_{\text{tot}} = \sqrt{\frac{1}{N} \sum_{n=1}^N \Delta P_n^2} = \sqrt{\mu^2 + \sigma^2}.$$

$$\text{SNR} = \frac{\max(|\text{Signal}|)}{\max(|\text{Noise}|)}$$

- Metrics change for a particular signal scale and SNR
- So normally define these in a way that they can be compared to relative polarization error as we do in experiments.
- Basic metrics
  - Precision: Width of the distribution of similar scale polarization trials
  - Accuracy: bias, or shift away from true
  - Relative error is total/scale
  - Quoted goal of less than 1% but normally we plan a goal of 0.05% at the 5% polarized level...

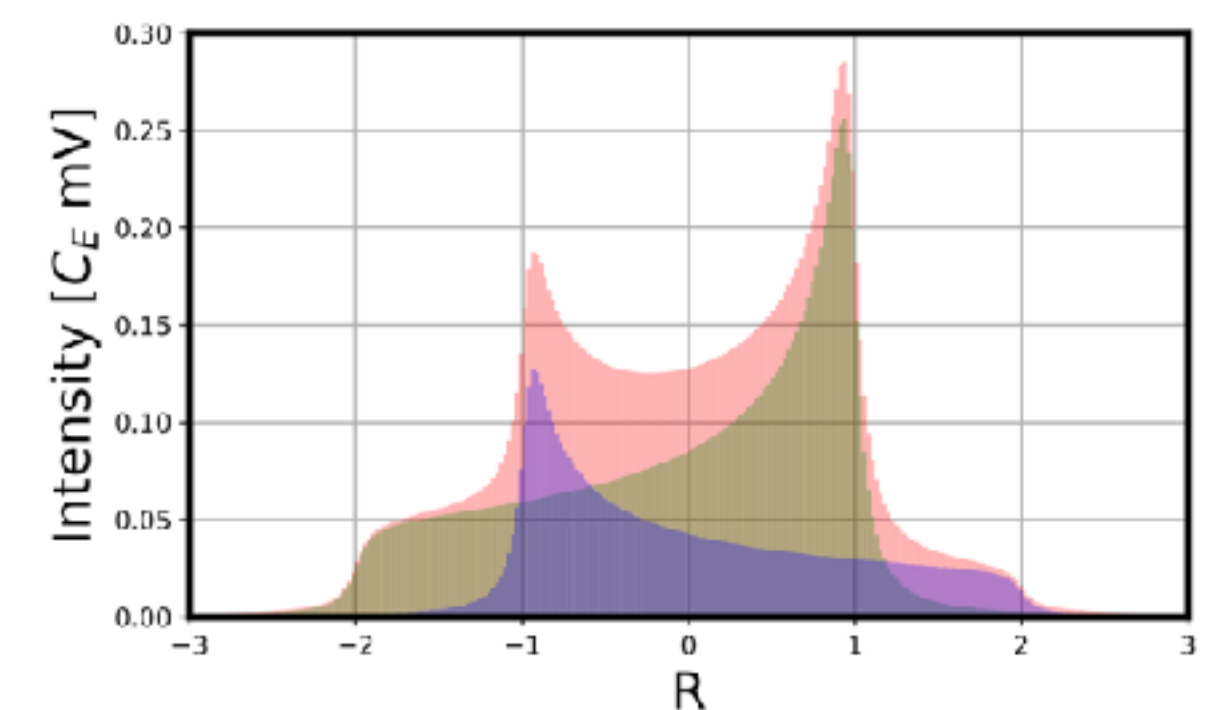
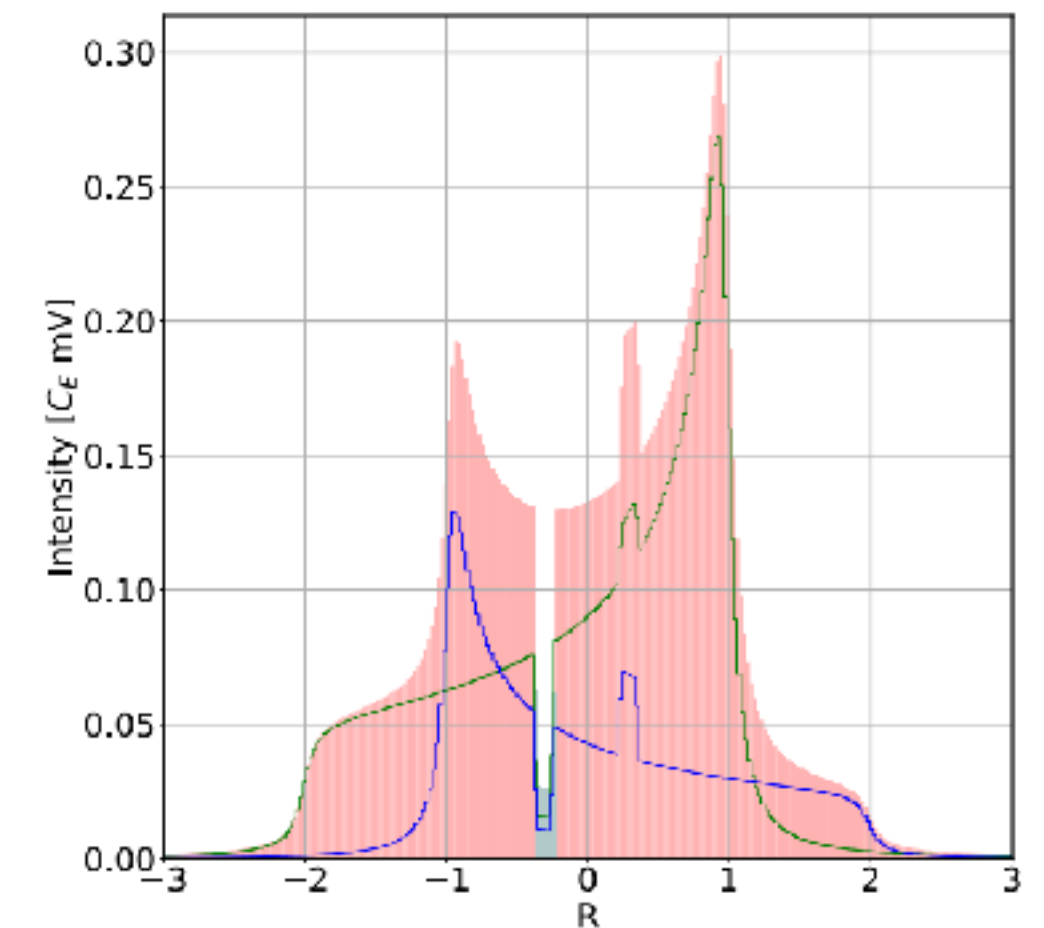
A confidence interval is a property of the repeated sampling behavior

# Deployment: Putting it all Together

## With realtime evolution

Differential binning, rates response (with spin diffusion) but no STC is used

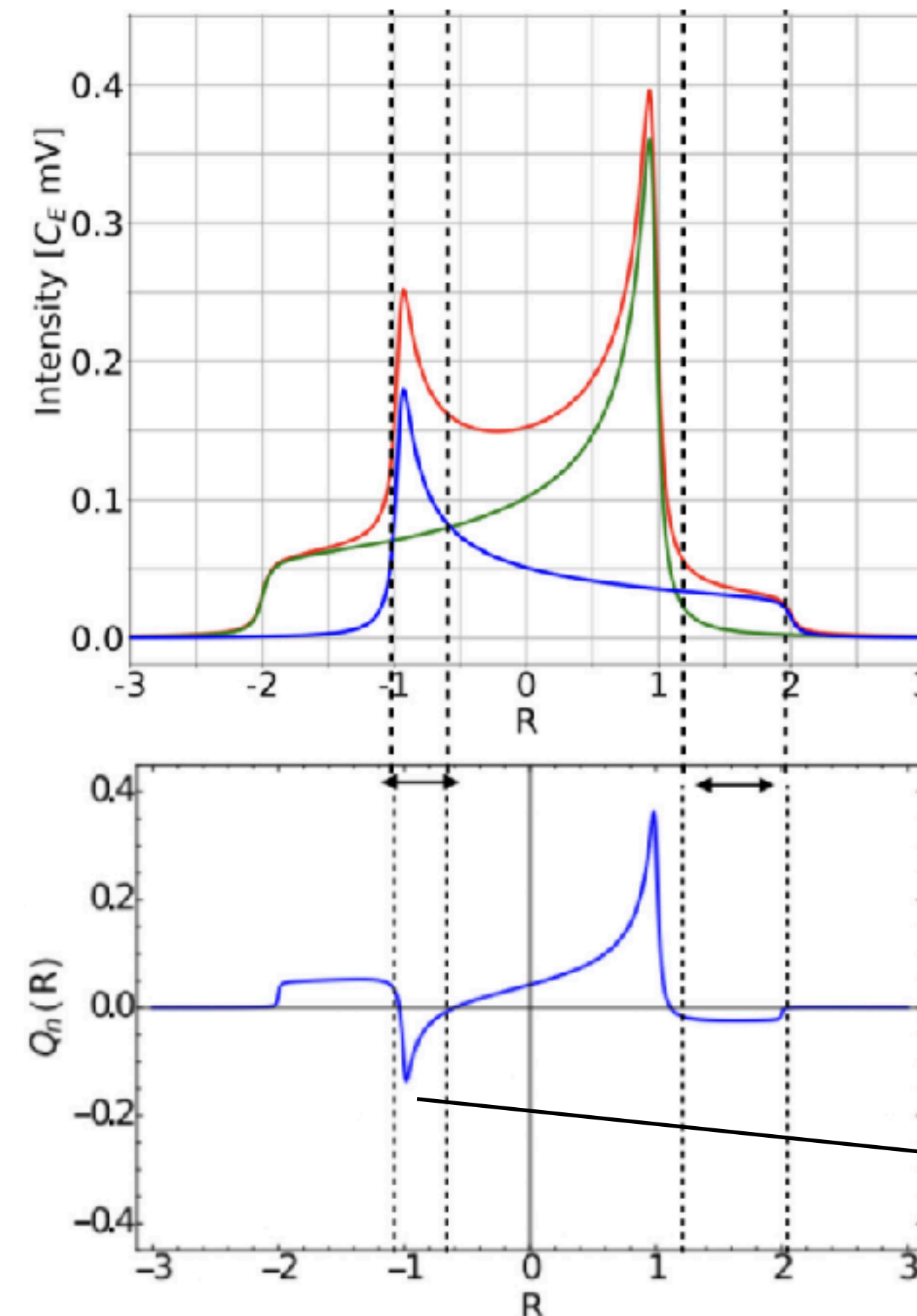
- Calibrate to interpret differential areas.
- Calibrate application of power sensitive pulses.
- Requires universal definition of Gamma and  $t_{\text{RF}}$ .
- High power, short pulses, narrow width (for shaping)
  - HP-limit is not required but HP provides best enhancement
- So best measurement, immediately after pulse, very fast!!
- Can feed either experimental data or MC data.



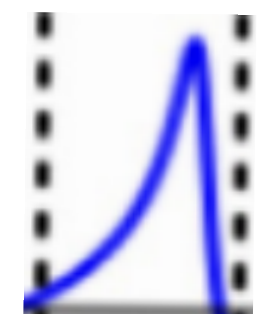
# How to Actually Maximize

## Assuming very narrow RF application

- Apply ss-RF as a function of power profile defined by  $Q(R)$  in the sweep pulses in the high power limit using the AFP coil rather than specialized NMR.
- Have fast DAQ to do high averaging of many sweeps quickly.
- Update sequence per bin and power profile as system evolves under DNP and radiation damage.
- Whats needed: Optimized coil (for both AFP and ss-RF), amplifier, generator, maybe upgrade to NMR DAQ, automated software to adaptively control power profile and pulse sequences.



Applied Power



# Area Extraction (Spin-1/2)

• Only looking at TE-Scale

Low Noise:  $N(0, \sigma = 2.55 \times 10^{-5})$

High Noise:  $N(0, \sigma = 2 * 2.55 \times 10^{-5})$

Typical scenario for Chi^2 polynomial subtraction  
+ Riemann sum: ~2-3%

| Method      | Noise | $\mu$ (%) | $\sigma$ (%) | $\delta P/P$ | SNR Range |
|-------------|-------|-----------|--------------|--------------|-----------|
| MLP         | Low   | 0.00128   | 0.00282      | 0.37%        | 2 – 354   |
|             | High  | 0.00127   | 0.00280      | 0.36%        | 1 – 89    |
| Riemann Sum | Low   | −0.05877  | 0.04278      | 16.79%       | 2 – 354   |
|             | High  | −0.05313  | 0.07711      | 15.18%       | 1 – 89    |

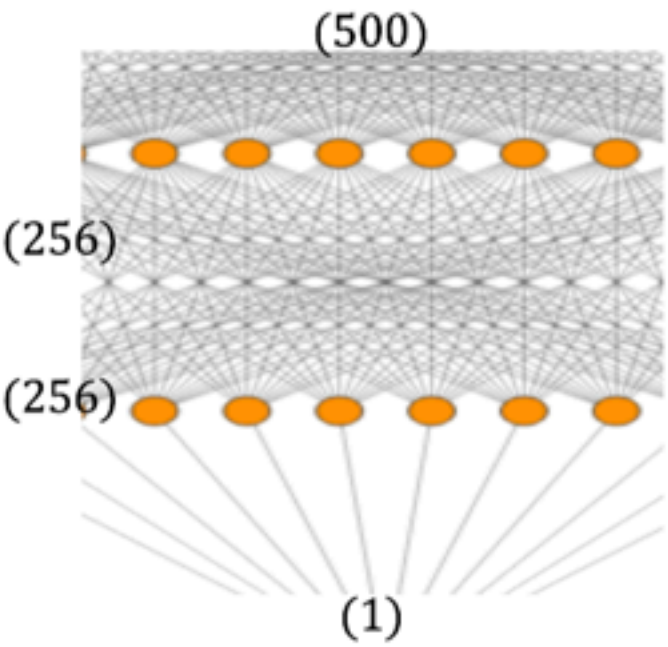
# Polarization Extraction (Spin-1)

| Range    | Noise | $\chi^2$ Fit Res. |          | $\delta P/P$ | DNN Res. |          | $\delta P/P$ | SNR Range |
|----------|-------|-------------------|----------|--------------|----------|----------|--------------|-----------|
|          |       | $\mu$             | $\sigma$ |              | $\mu$    | $\sigma$ |              |           |
| 10 – 15% | Low   | 0.327             | 0.901    | 2.725%       | 0.010    | 0.061    | 0.080%       | 51 – 114  |
|          | High  | 0.362             | 1.832    | 3.017%       | 0.015    | 0.063    | 0.120%       | 28 – 59   |
| 20 – 25% | Low   | 0.260             | 0.646    | 1.156%       | 0.006    | 0.055    | 0.027%       | 88 – 202  |
|          | High  | 0.266             | 1.297    | 1.182%       | 0.009    | 0.056    | 0.040%       | 54 – 104  |
| 30 – 35% | Low   | 0.248             | 0.572    | 0.763%       | 0.003    | 0.051    | 0.009%       | 137 – 283 |
|          | High  | 0.237             | 1.146    | 0.729%       | 0.004    | 0.057    | 0.012%       | 80 – 155  |
| 40 – 45% | Low   | 0.259             | 0.533    | 0.609%       | 0.002    | 0.052    | 0.005%       | 235 – 411 |
|          | High  | 0.244             | 1.063    | 0.574%       | 0.003    | 0.060    | 0.007%       | 123 – 204 |
| 50 – 55% | Low   | 0.259             | 0.501    | 0.493%       | 0.001    | 0.065    | 0.002%       | 306 – 512 |
|          | High  | 0.241             | 0.998    | 0.459%       | 0.001    | 0.065    | 0.002%       | 149 – 251 |

Low Noise:  $N(0, \sigma = 2.55 \times 10^{-5})$

High Noise:  $N(0, \sigma = 2 * 2.55 \times 10^{-5})$

Recall: Dulya's Fit error: ~3%



1. Reduce extraction errors
  - Baseline subtraction
  - Polynomial subtraction
  - Lineshape fitting
2. Achieve reliable results even with large range of noise and shifts
3. Develop flexible software that improves reliability even when working Q-meter outside its design specifications
4. Enable *real-time* polarization readings
5. Apply to ss-RF/AFP type manipulations

# Polarization Extraction (Spin-1)

TE-Scale  
SNR ~ (5.2 × 10<sup>−5</sup>, 21.5)

| Model         | $\Delta P/P$ |
|---------------|--------------|
| CNN           | 3.8%         |
| MLP           | 8.2%         |
| $\chi^2$ -fit | 16%          |

Recall: Dulya's total error for enhanced TE: ~5%

Input

Multi-Scale Conv

Residual Block

⋮

Residual Block

⋮

Global Pooling

Weight Layer

Output

$x$

$u$

$F_{in}(C)$

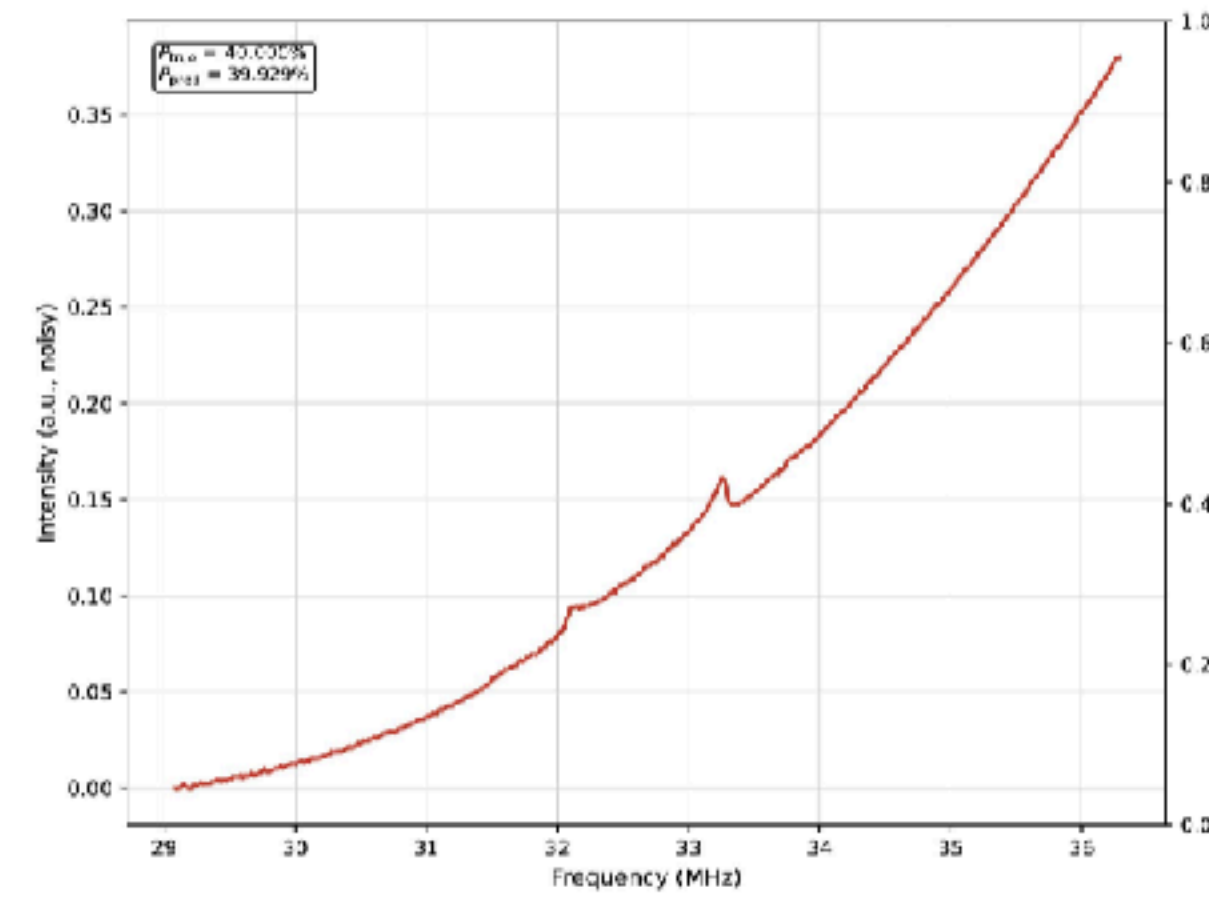
$F_{in}(C; W)$

$F_{scale}(C; T)$

$x$

With addition of Squeeze-and-Excitation Block

arXiv:1709.01507

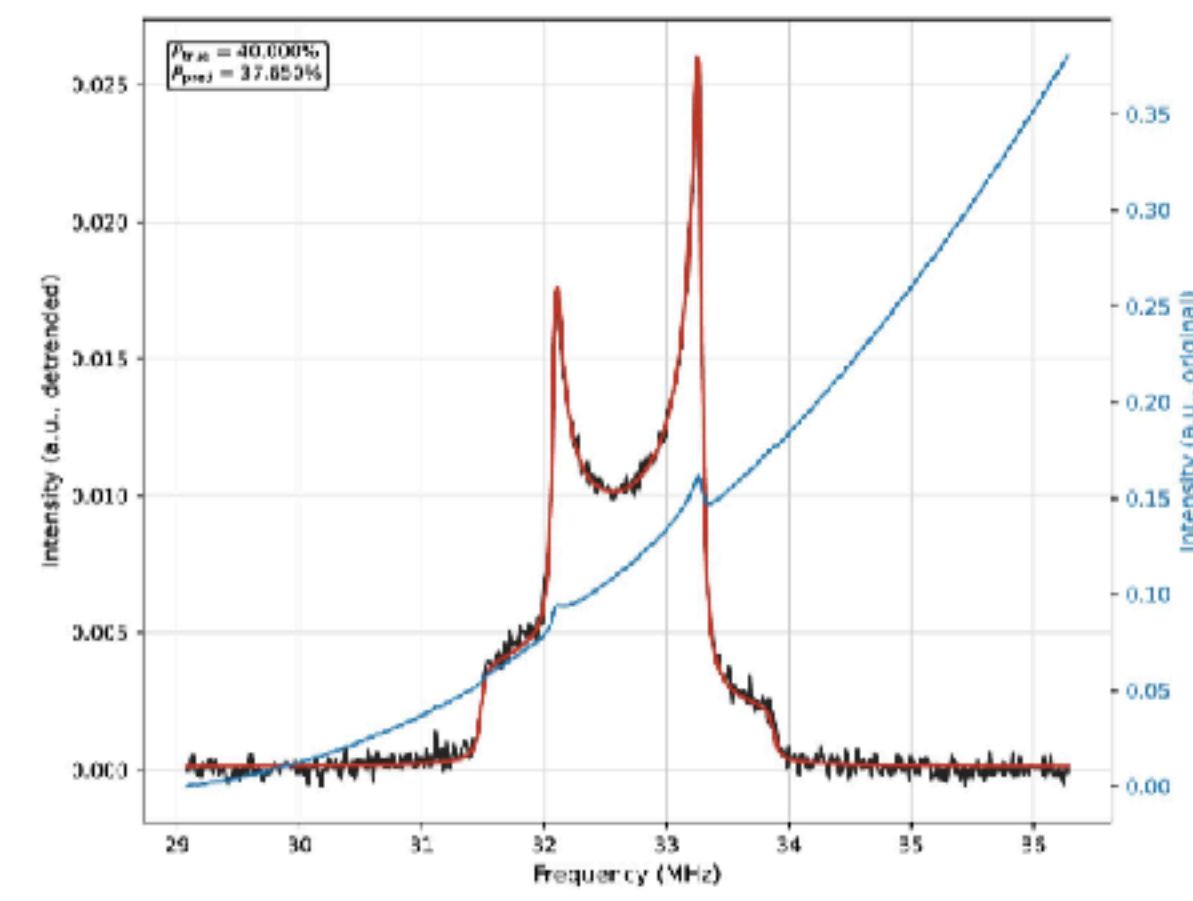


$$P_{True} = 40\%$$

$$P_{Pred} = 39.929\%$$

$$\Delta P/P = 0.18\%$$

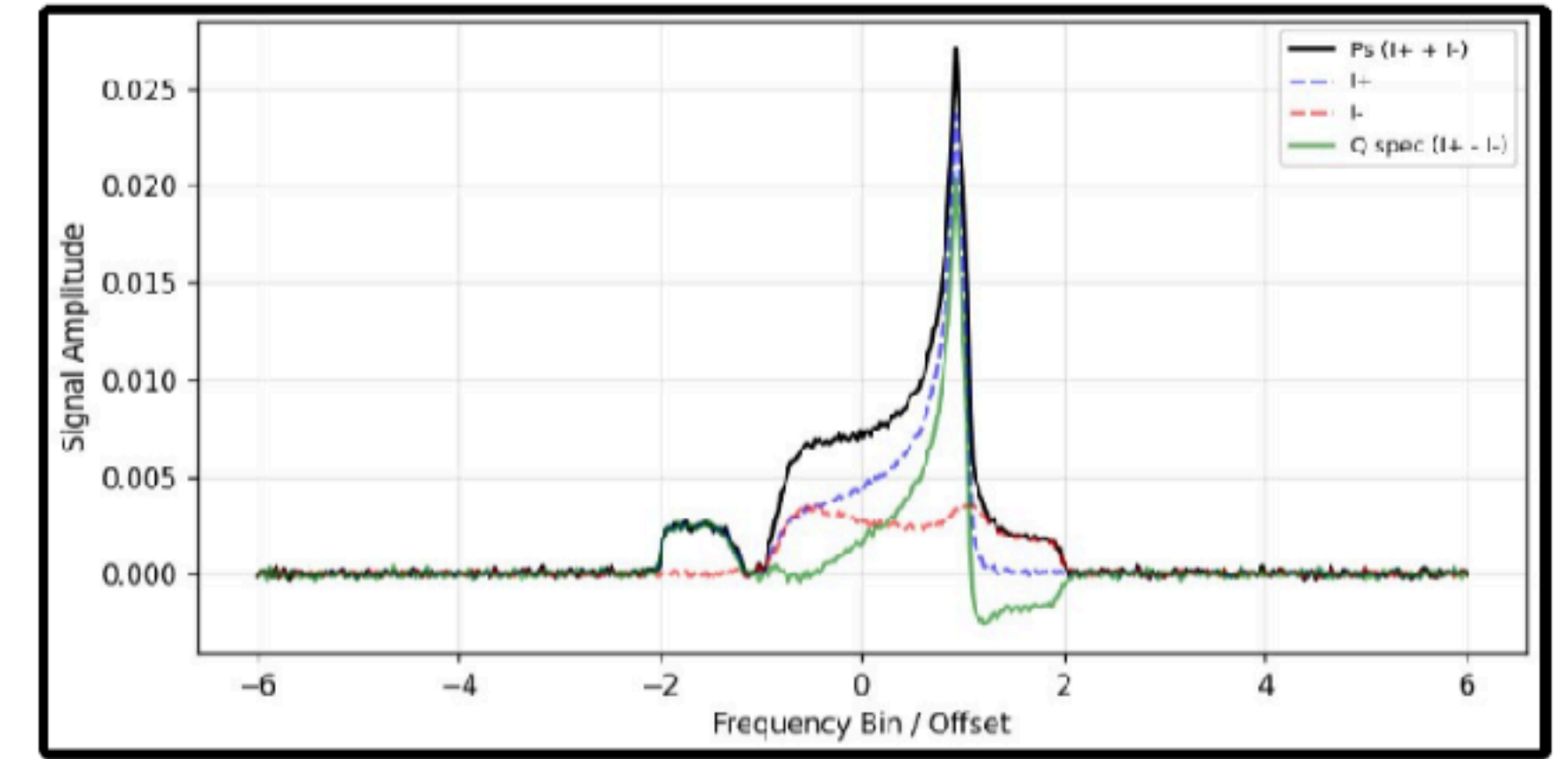
$SNR = 105$



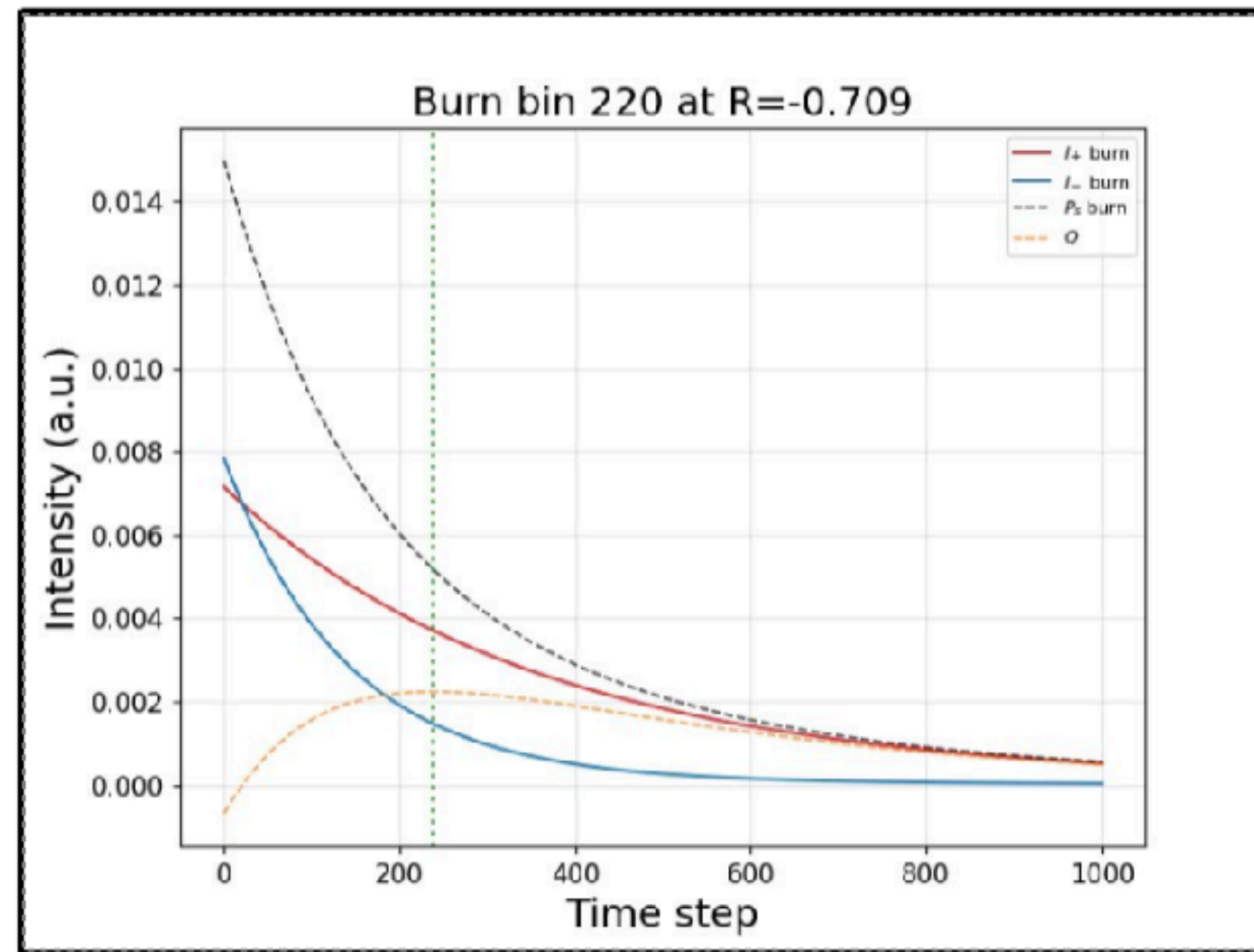
$$P_{True} = 40\%$$

$$P_{Pred} = 37.650\%$$

$$\Delta P/P = 5.88\%$$

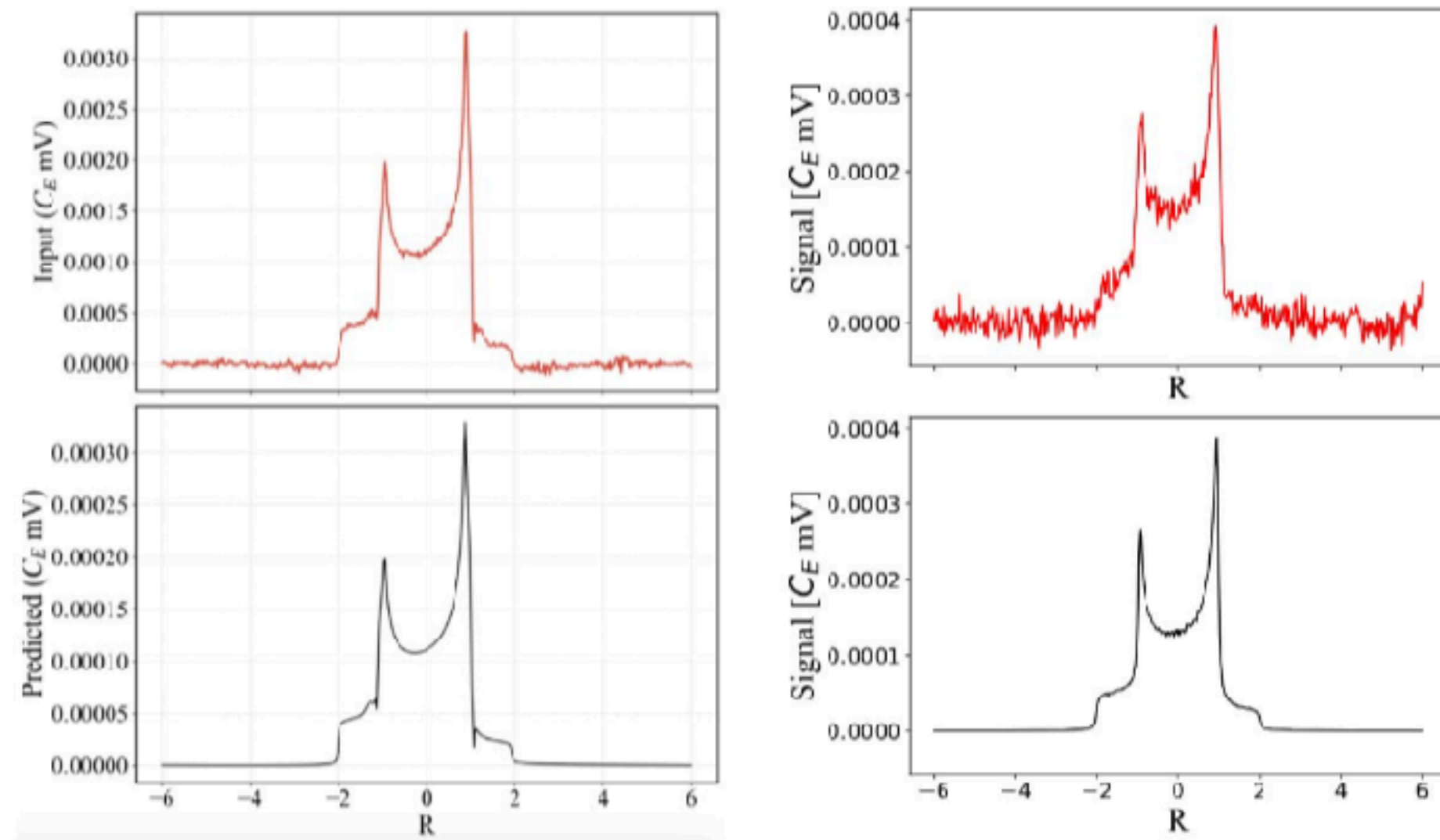


| Range  | P Res.   |          | Q Res. |          |
|--------|----------|----------|--------|----------|
|        | $\mu$    | $\sigma$ | $\mu$  | $\sigma$ |
| 15–20% | 0.00006  | 0.00009  | 0.180  | 0.215    |
| 20–25% | 0.00001  | 0.00007  | 0.161  | 0.138    |
| 25–30% | 0.00004  | 0.00016  | 0.099  | 0.100    |
| 30–35% | 0.00005  | 0.00010  | 0.081  | 0.052    |
| 35–40% | −0.00003 | 0.00011  | 0.075  | 0.050    |
| 40–45% | −0.00041 | 0.00019  | 0.161  | 0.133    |
| 45–50% | −0.00026 | 0.00019  | 0.125  | 0.086    |
| 50–55% | −0.00043 | 0.00029  | 0.159  | 0.112    |
| 55–60% | −0.00011 | 0.00040  | 0.080  | 0.075    |
| 60–65% | −0.00007 | 0.00000  | 0.121  | 0.000    |



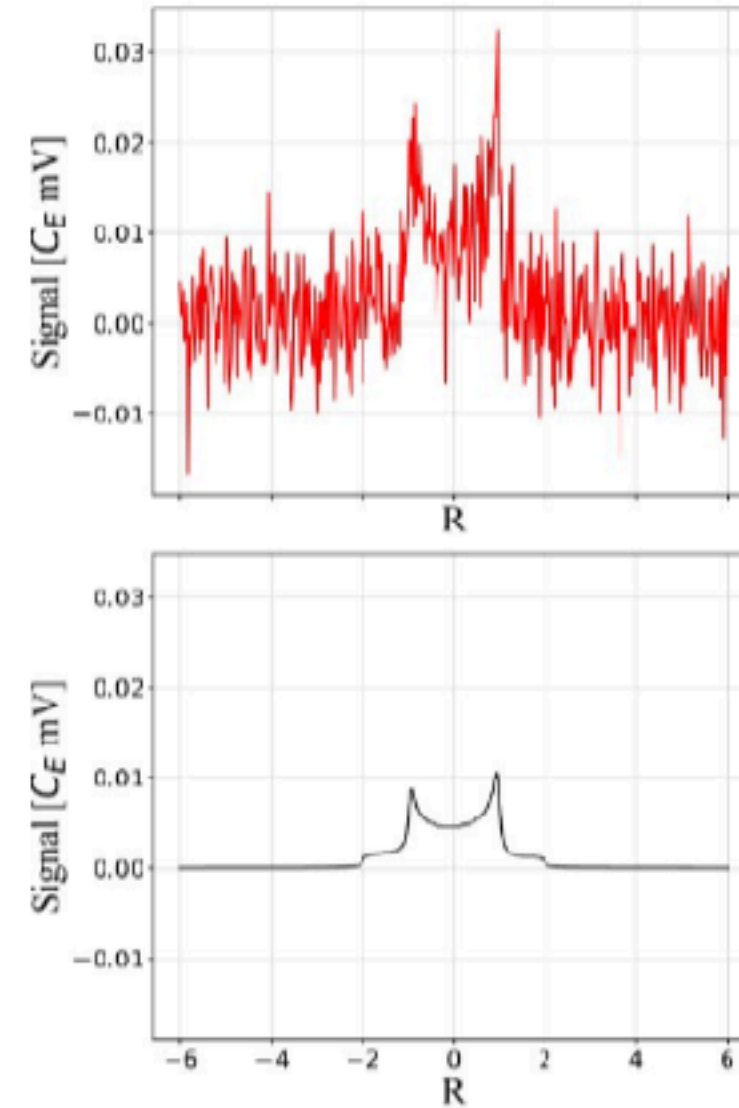
# Denoising Autoencoder

*Eur.Phys.J.A* 62 (2026) 153

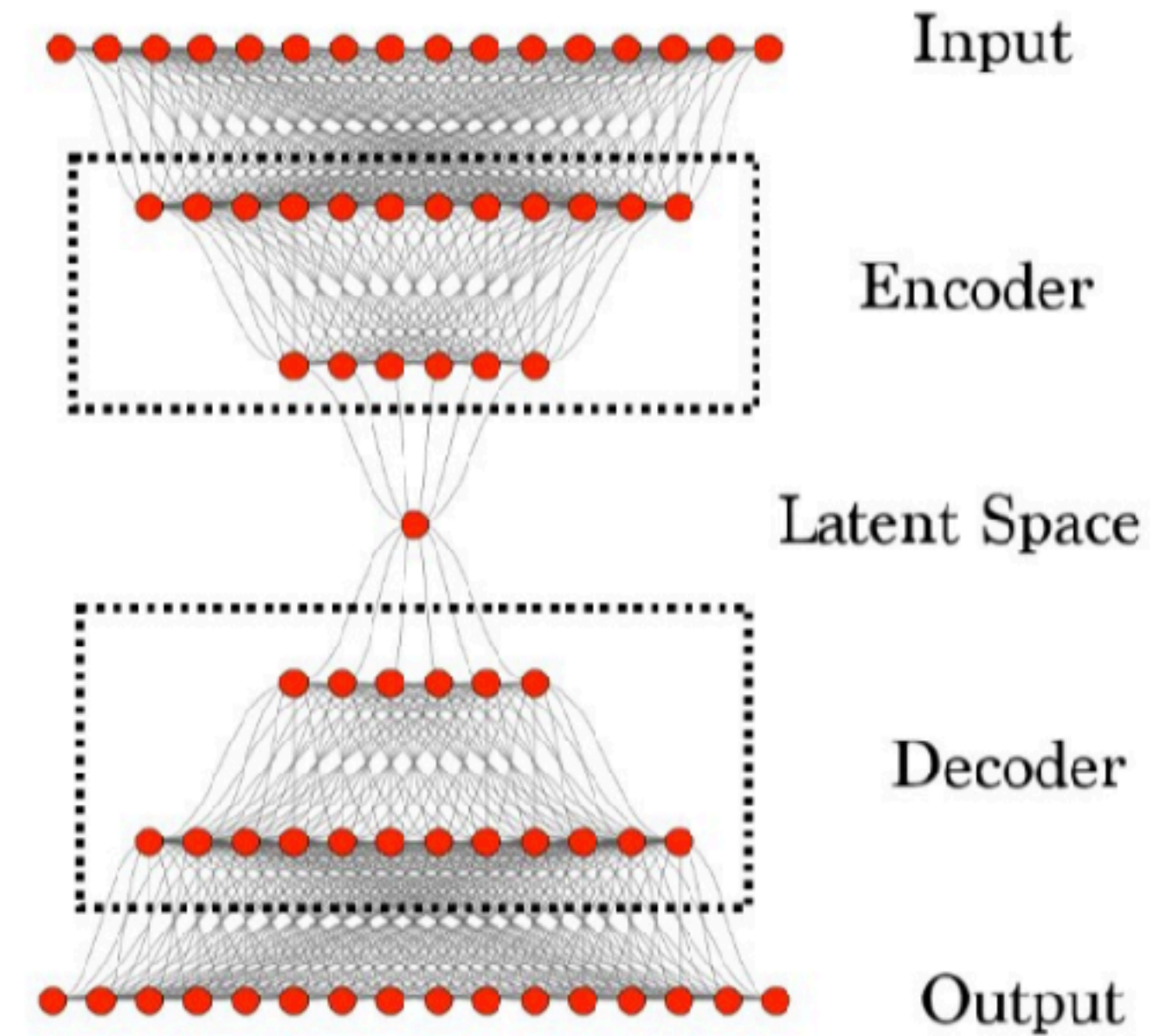


Experimental

$$P_{\text{true}} = 5\% \quad \text{SNR} = 0.805$$



$$\begin{aligned} \mu &= 5 \times 10^{-8} \\ \sigma &= 1.5 \times 10^{-6} \\ P_{\text{pred}} &= 4.98 \pm 0.04\% \end{aligned}$$



# Hardware Plan for UVA

## Use just basic set up

- Measure optimize profile with spectrum analyzer so we know we have the right one before claiming is transferable to another lab.
  - Not able to measure overlap without power and timing
  - Essential proof of principle requires the coupling between lineshape and power profile as measured within the spectrum
  - For real prototype you need both fast application and acquisition synchronized to configurations of pulse sequences for both measurement and shaping (can't yet speak to how to do this with the planned limitations)